

Experiment 5.4

Determination and plotting of Potential Difference with both r1 and r2 varying

Code File Name: Exp5_4_Both_r1_r2_Varying.m

Code

```
clc;
epsilon=8.854e-12;
k=1/(4*pi*epsilon);
q=1e-6;
r1=linspace(0.5,5,100);
r2=linspace(5,10,100);
V=k*q*(1./r1 - 1./r2);
plot(r1,V,'LineWidth',2);
xlabel('r1 (m)');
ylabel('Potential Difference (V)');
title('Both r1 and r2 varying');
```

Output Graph

